HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

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FUNDAMENTALS OF OPTIMIZATION

Introduction

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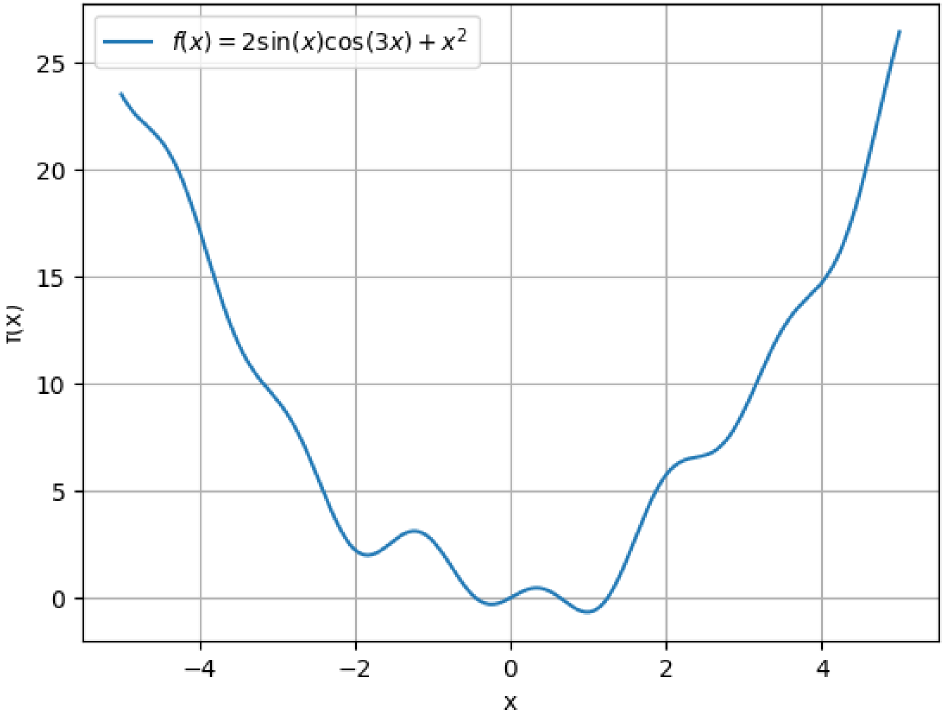
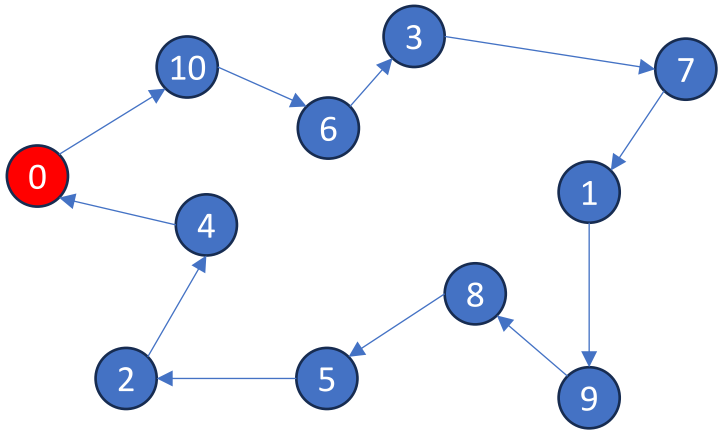
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Maximize or minimize some function relative to some set (range of choices)
- The function represents the quality of the choice, indicating which is the “best”
- Example
  - Minimize $f(x) = 2\sin(x)\cos(3x) + x^2$
  - A shipper need to find the shortest route to deliver packages to customers 1, 2, …, N





- $x \in R^n$: vector of decision variables x_j, for $j = 1, 2, \ldots, n$
- $f : R^n \rightarrow R$ is the objective function
- $g_i : R^n \rightarrow R$ is the constraint function defining restriction on x, $i = 1, 2, \ldots, m$

minimize $f(x)$ over $x = (x_1, x_2, \ldots, x_n) \in X \subset R^n$ satisfying a property P:

$$g_i(x) \le b_i,\ i = 1, 2, \ldots, s$$
$$g_i(x) = d_i,\ i = s + 1, 2, \ldots, m$$

- Example

$$
\begin{aligned}
\min f(x) = {} & 3x_1 - 5x_2 + 10x_3 \\
& x_1 + x_2 + x_3 \leq 10 \\
& 2x_1 + 4x_2 - 5x_3 = 9 \qquad \text{(Linear Program)} \\
& x_1, x_2 \in R^+, x_3 \in Z
\end{aligned}
$$

$$
\begin{aligned}
\min f(x) = {} & 4x_1^2 + 3x_2^2 - 7x_1 x_3 \\
& x_1 + x_2^3 + 4x_3 \leq 10 \\
& 2 x_1^2 + 4x_2 - 5x_3 = 7 \qquad \text{(Nonlinear Program)} \\
& x_1, x_2 \in R^+, x_3 \in Z
\end{aligned}
$$

INTRODUCTION TO OPTIMIZATION PROBLEMS

- General optimization problems
  - Very difficult to solve
- Some special cases can be solved easily (well-known algorithms)
  - Linear programming
  - Least square problem
  - Some shortest path problems on networks
  - Etc.

INTRODUCTION TO OPTIMIZATION PROBLEMS

- Classification
  - Linear Programming (LP): f and g_i are linear
  - Nonlinear Programming (NLP): some function f, g_i are nonlinear
  - Continuous optimization: f and g_i are continuous on an open set containing X, X is closed and convex
  - Integer Programming (IP): $X \subseteq \{0,1\}^n$ or $X \subseteq Z^n$
  - Constrained optimization: $m > 0$, $X \subset R^n$
  - Unconstrained optimization: $m = 0$, $X = R^n$
- Combinatorial optimization:
  - Solutions can be an arrangement, a permutation, a subset of a given set
  - Solution space is finite
  - Routing, Bin Packing, Scheduling, Timetabling, Coloring
- A combinatorial optimization problem is a discrete optimization problem, but some discrete optimization problems are not combinatorial optimization
  - Example:

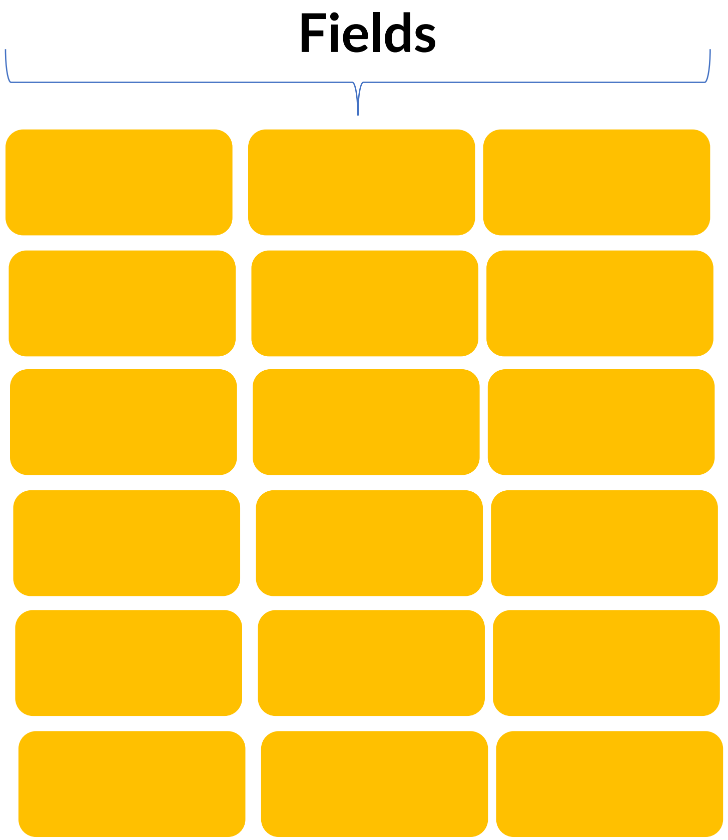
$$\min f(x_1, x_2, x_3) = 4x_1^2 + 3x_2^2 - 7x_1 x_3$$
$$x_1 + x_2^3 + 4x_3 \leq 10$$
$$2x_1^2 + 4x_2 - 5x_3 = 7$$
$$x_1, x_2, x_3 \in Z$$

Discrete optimization but not combinatorial optimization

- Applications
    - Production Planning
    - Routing in transportation
    - Scheduling
    - Assignment
    - Packing
    - Timetabling
    - Network designs
    - Machine learning
    - . . .

- Production Planning

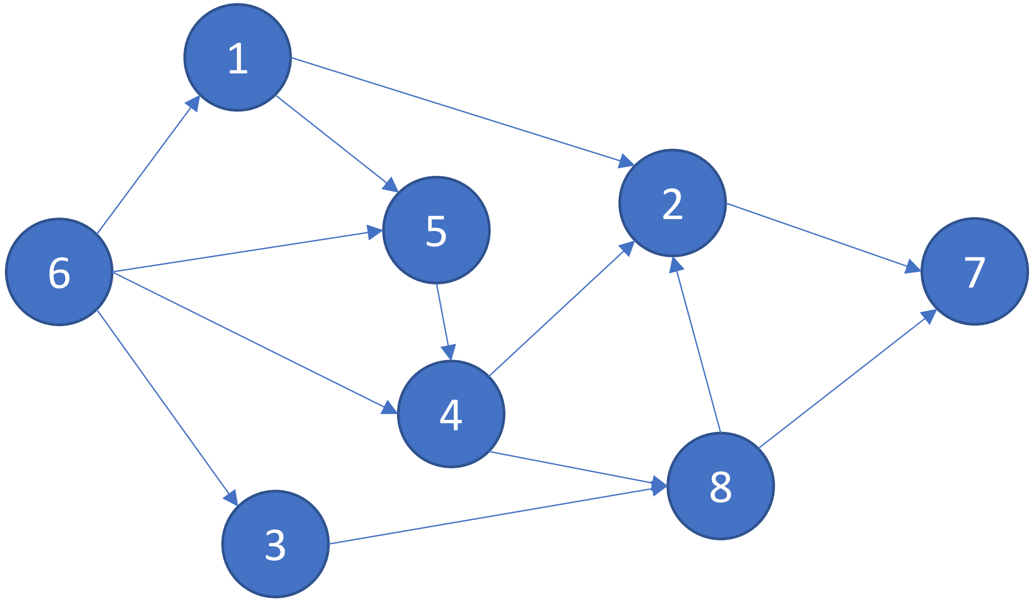
SKU	Chart	Demand
		10000
		25000
		32000
		42500

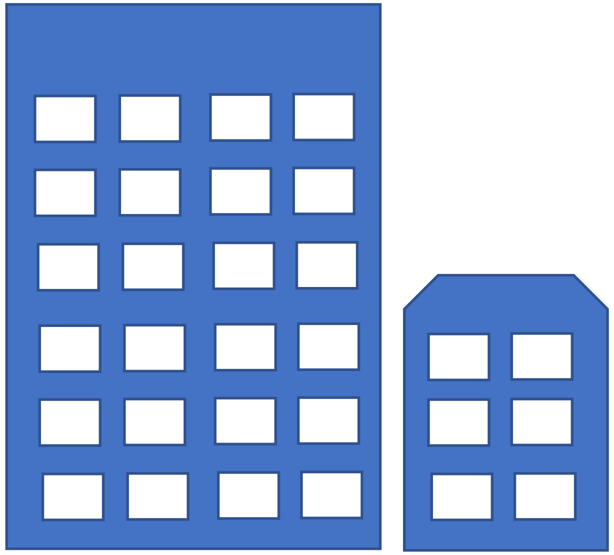


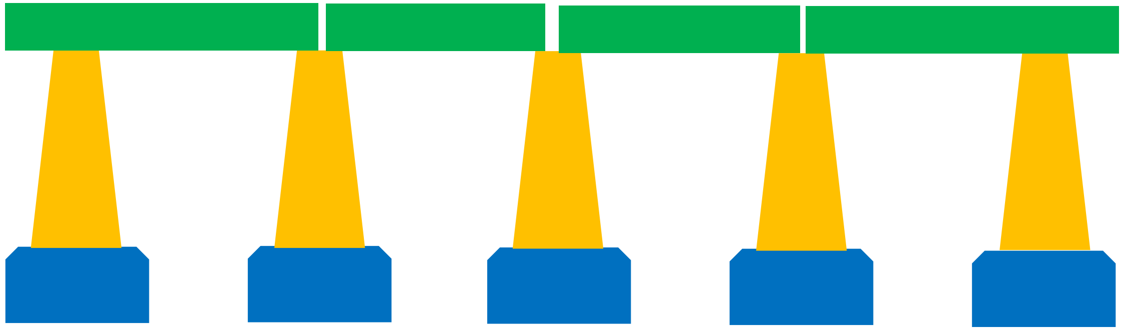
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Construction Planning

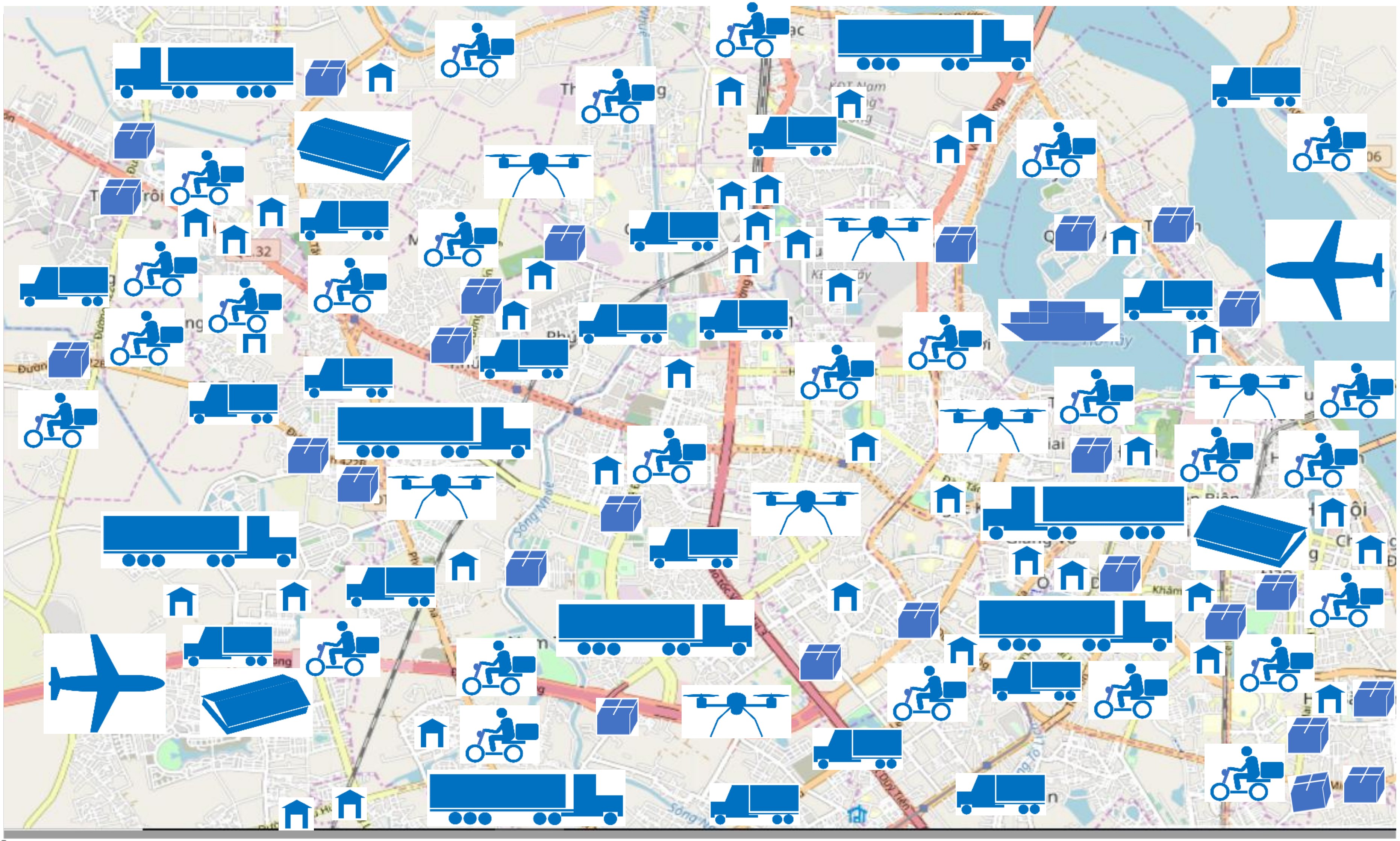
Task	Duration	Predecessors
1	30	6
2	20	1,4,8
3	15	6
4	25	5,6
5	20	1,6
6	45	
7	40	2,8
8	30	3,4



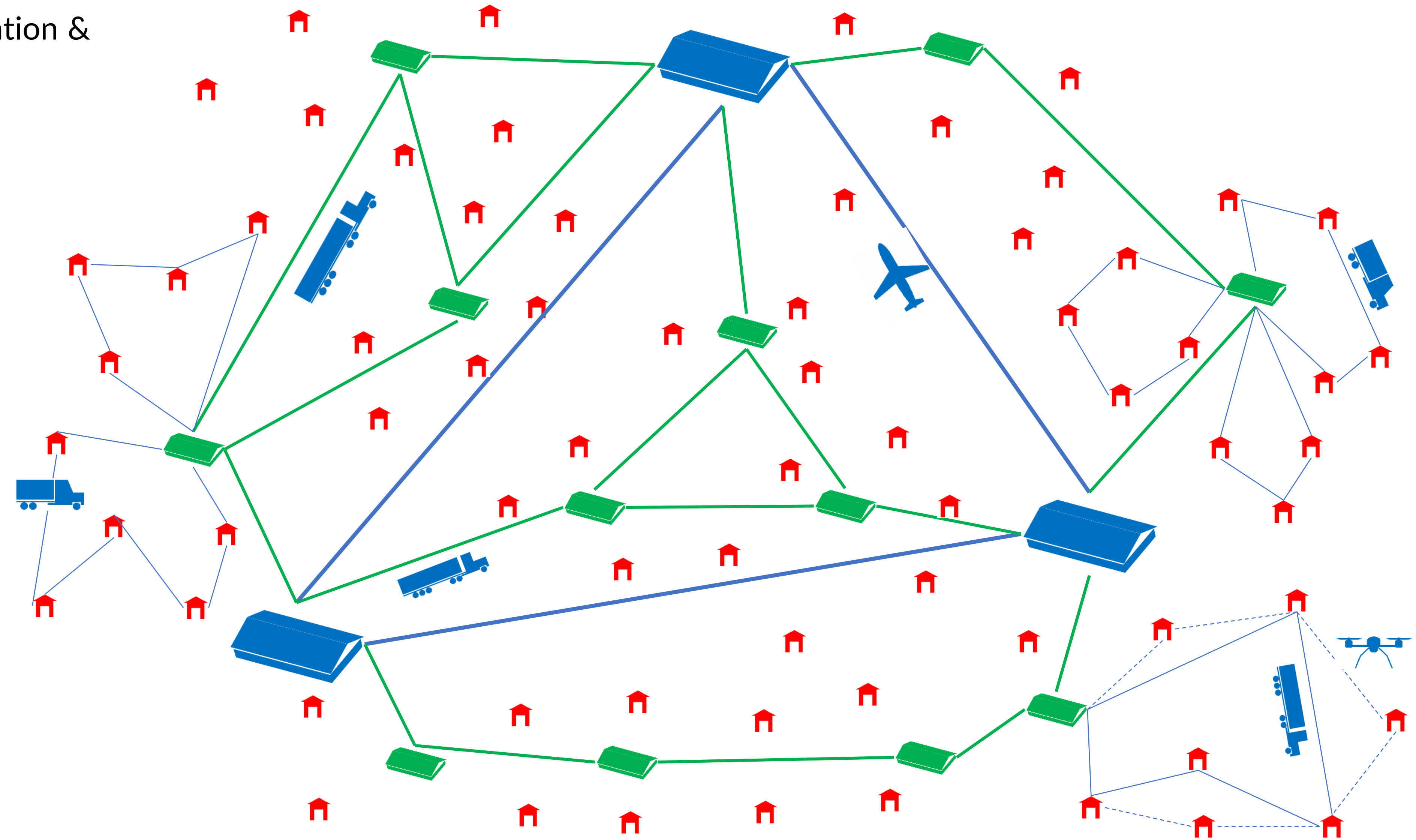




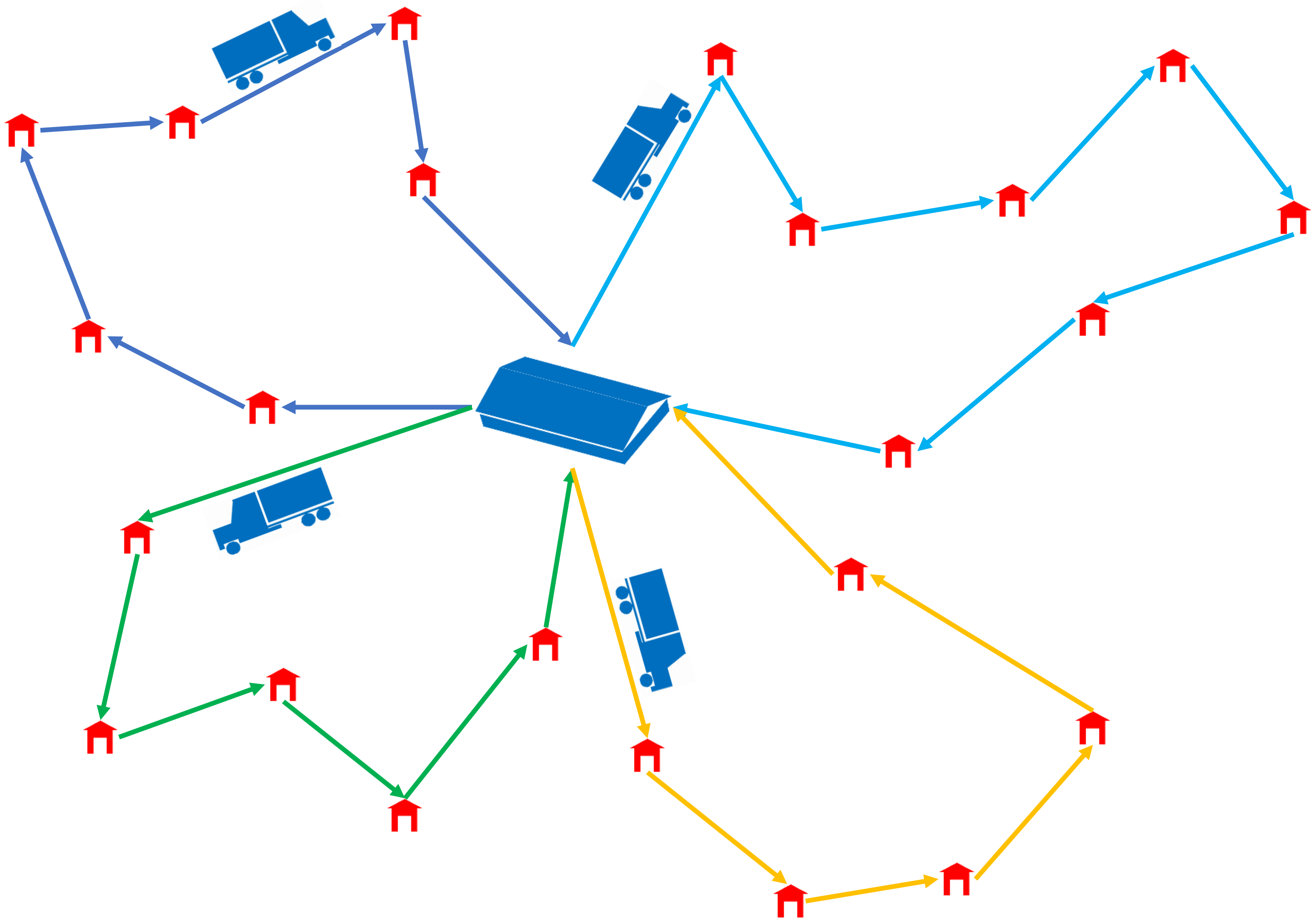
- Routing in transportation & logistics



- Routing in transportation & logistics



- Routing in transportation & logistics
    - How to make a route plan for delivering goods to customers from a central depot?



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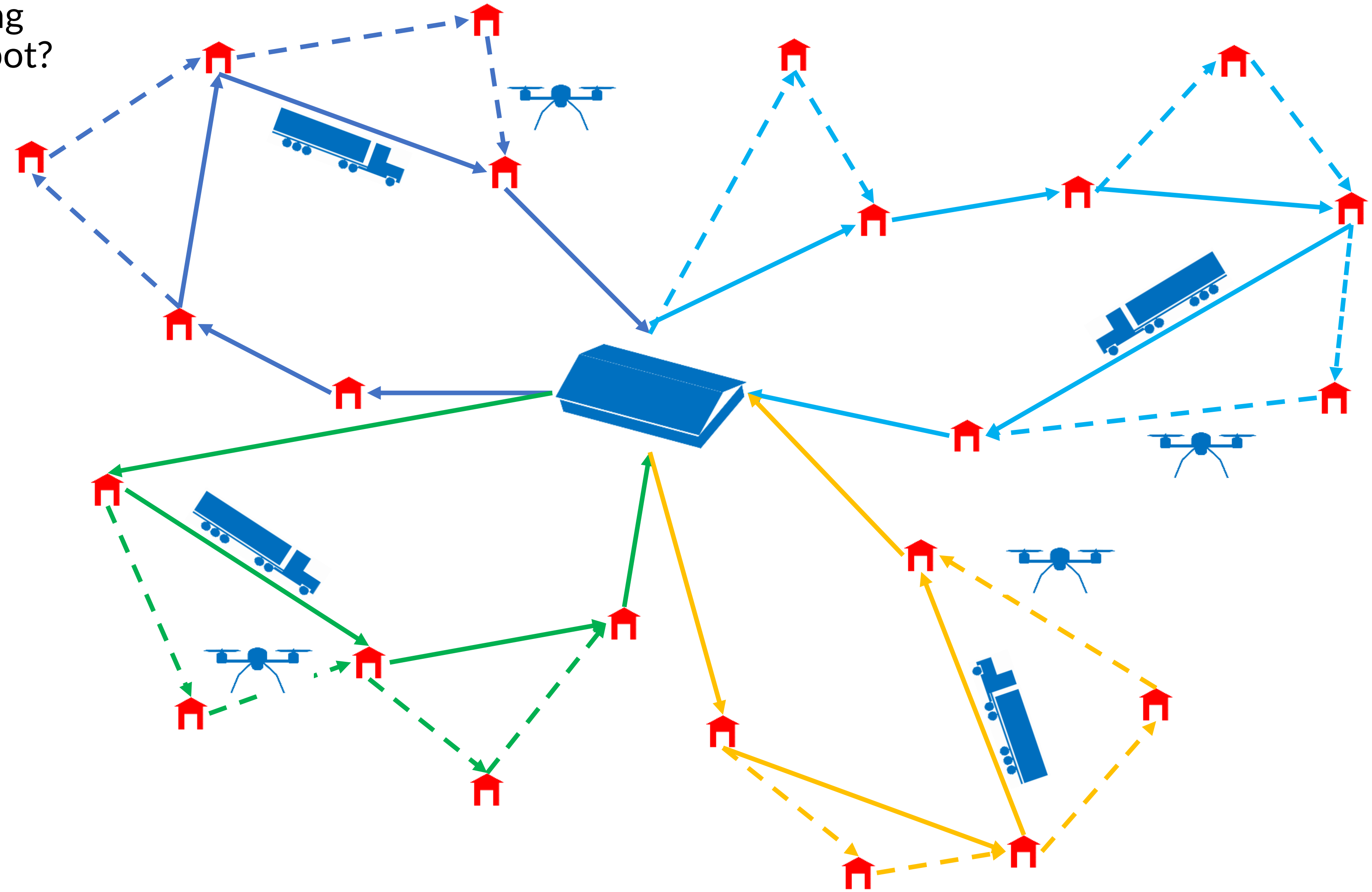
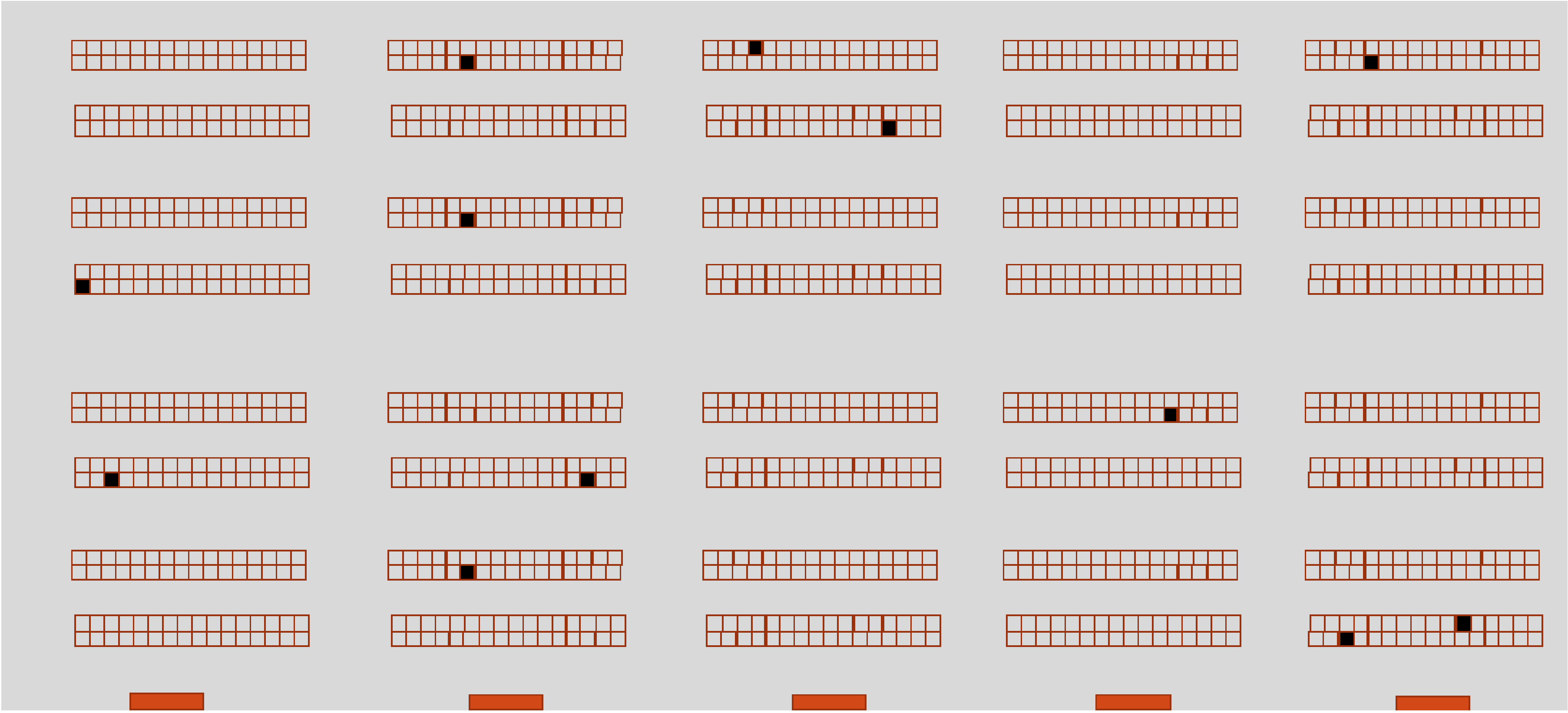


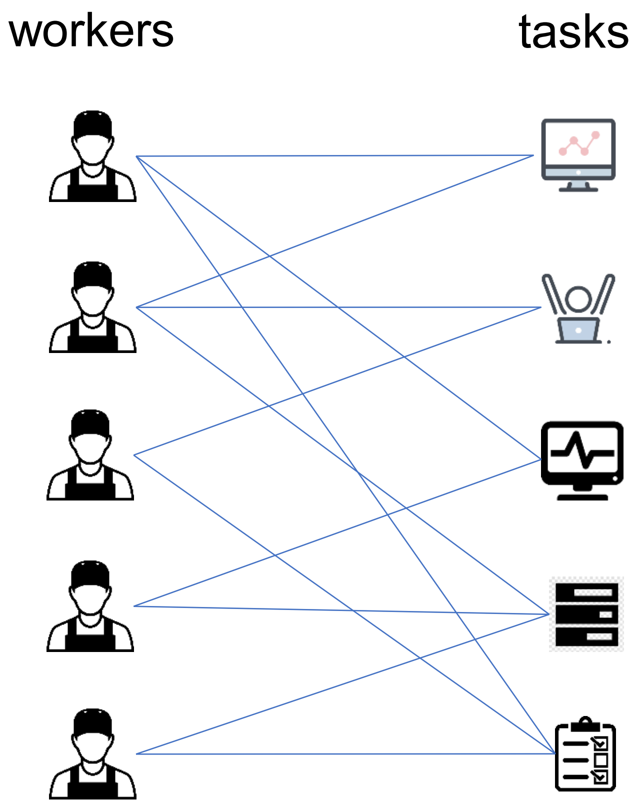
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Routing in transportation & logistics
  - How to make a route plan for picking up items in a very large warehouse?



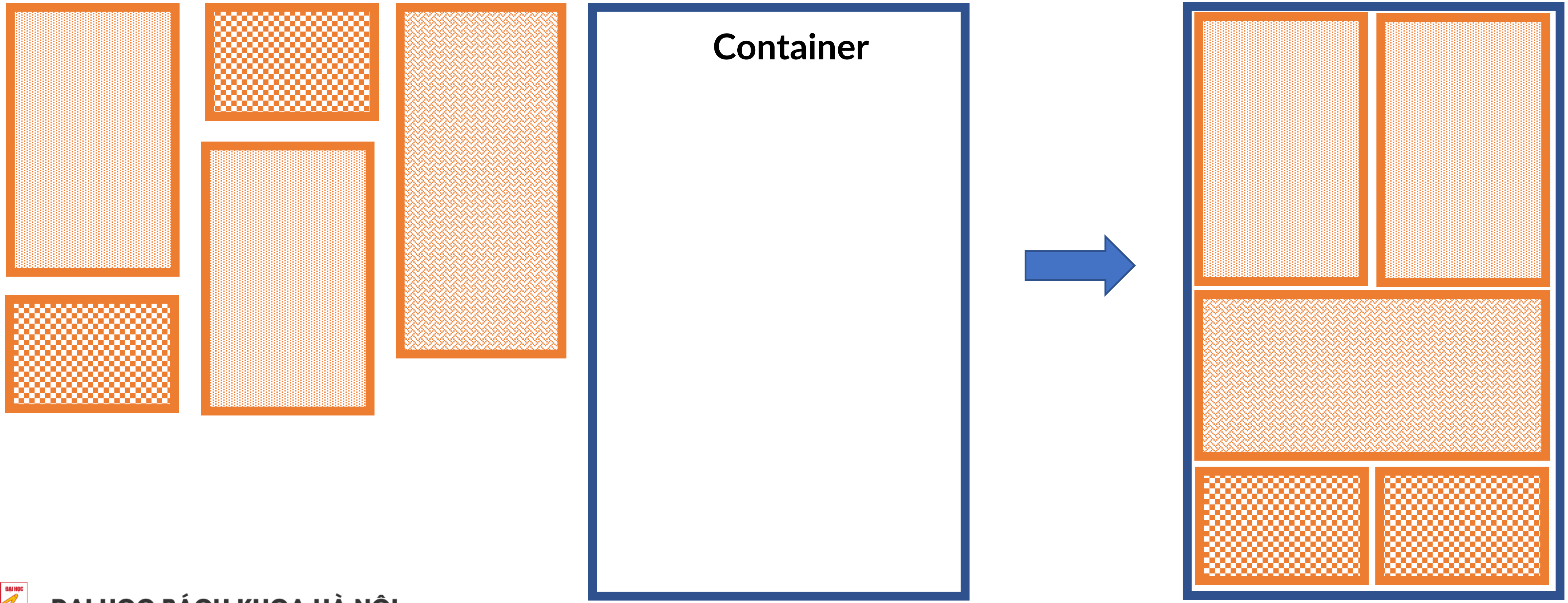
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Assignment
  - How to assign tasks to workers in an optimal way ?



4		6		8
2	6		7	
	5			6
		1	4	
			6	3

- Packing
    - How to arrange items in a container in an optimal ways?



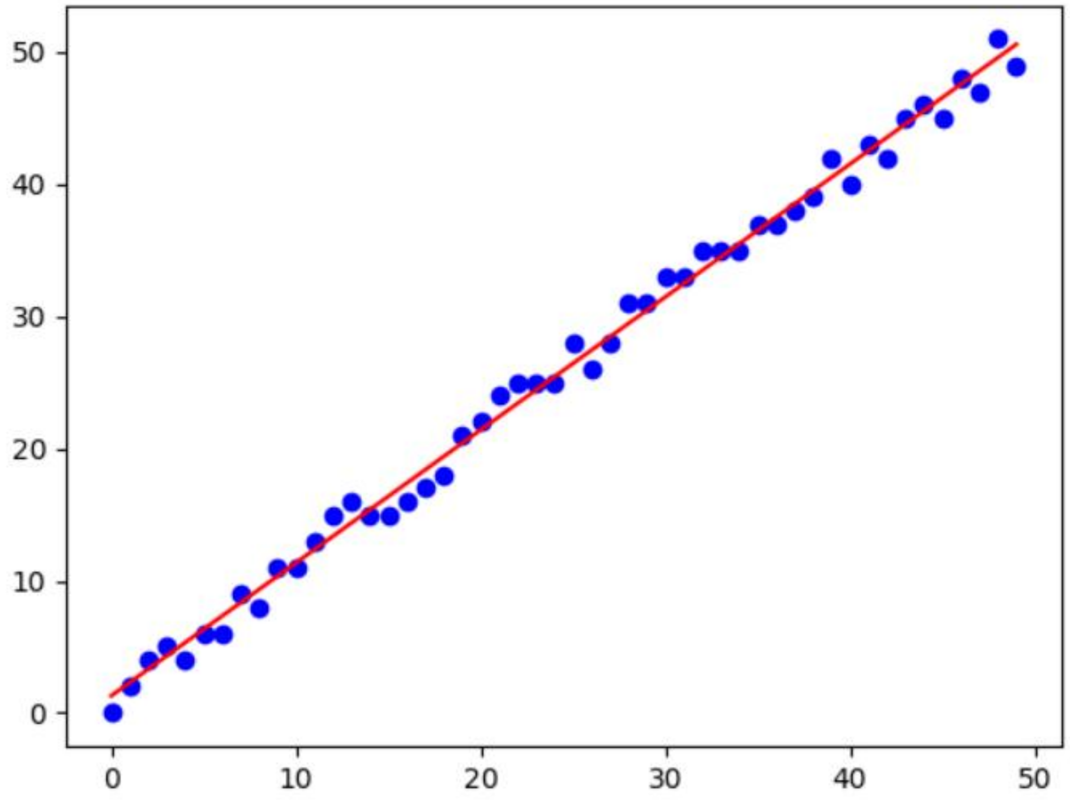
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Timetabling: How to arrange courses into time slots?
- [IT3011, 3, 100], [IT3020, 4, 120], [IT4663, 3, 60], [IT3220, 4, 90], [IT3030, 3, 100], [IT3170, 2, 120]

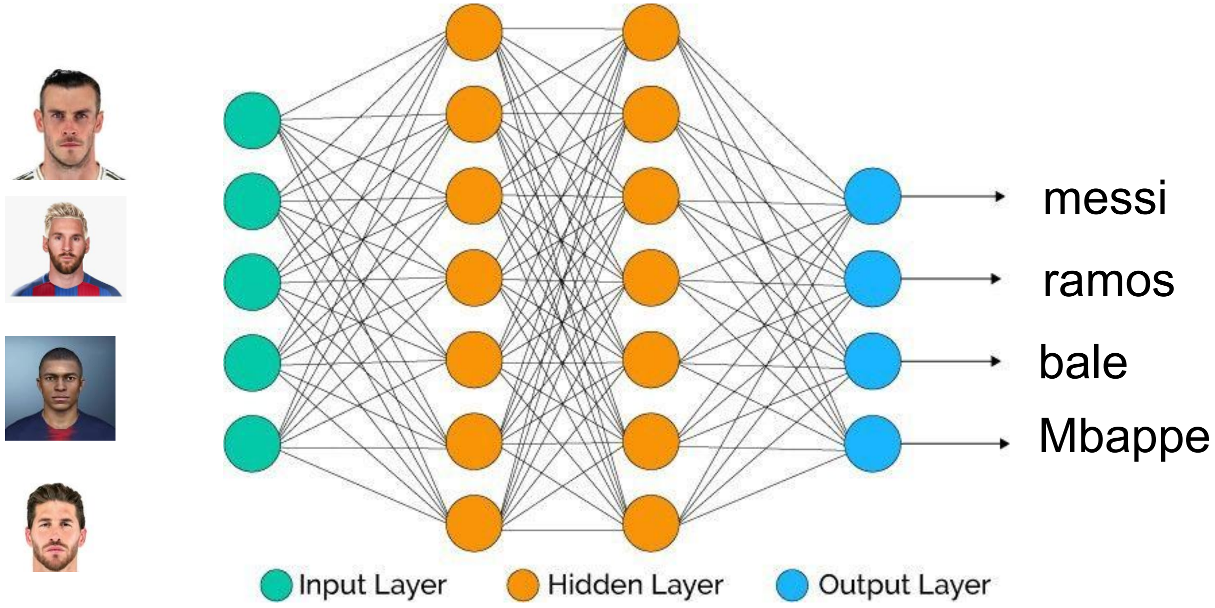
Room	Cap	Monday										Tuesday										Wednesday									
		Morning					afternoon					Morning					Afternoon					Morning					Afternoon				
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D9-101	150																														
C7-205	80																														
D5-301	120																														
D8-102	100																														

- Machine learning
  - Prediction

X	Y
43	45
44	46
45	45
46	48
47	47
48	51
49	49
50	?



- Machine learning
  - Computer Vision



MODELLING

- Transform the problem description to a mathematical model
  - Define variables and domains of variables
  - State constraints and objective functions

Problem description
A company must decide to make a plan to produce 2 products P1, P2. • The revenue received when selling 1 unit of P1 and P2 are respectively 5$ and 7$ • The manufacturing cost when producing P1 and P2 are respectively 5$ and 3$ • The storage cost in warehouses for 1 unit of P1 and P2 are respectively 2$ and 3$ Compute the production plan so that • Total manufacturing cost is less than or equal to 200$ • Total storage cost is less than or equal to 150$ • Total revenue is maximal

MODELLING

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<table>
<tr><th>Problem description</th><th>Mathematical model</th></tr>
<tr><td>A company must decide to make a plan to produce 2 products P1, P2.
• The revenue received when selling 1 unit of P1 and P2 are respectively 5$ and 7$
• The manufacturing cost when producing P1 and P2 are respectively 5$ and 3$
• The storage cost in warehouses for 1 unit of P1 and P2 are respectively 2$ and 3$
Compute the production plan so that
• Total manufacturing cost is less than or equal to 200$
• Total storage cost is less than or equal to 150$
• Total revenue is maximal</td><td>• Variable definition
  • X_1, X_2: amount of P1 and P2 to be produced
  • Domains: $X_1, X_2 \in$ R and nonnegative
• Constraints
  • $5X_1 + 3X_2 \leq 200$
  • $2X_1 + 3X_2 \leq 150$
• Objectives
  • Maximize $5X_1 + 7X_2$</td></tr>
</table>

MODELLING

- **Worker Task Assignment**
  - Given 2 tasks 1, 2 and 2 workers 1, 2. The cost of assigning a task to a worker is described by the matrix *c* below in which *c*(*i*, *j*) is the cost of assigning task *i* to worker *j* (*i*, *j* = 1, 2)
  - Find the assignment solution such that
    - Each task is assigned to exactly one worker
    - Each worker is assigned to exactly one task
    - The total cost is minimal

	1	**2**
1	4	2
2	7	1

- **Worker Task Assignment**
  - Given 2 tasks 1, 2 and 2 workers 1, 2. The cost of assigning a task to a worker is described by the matrix *c* below in which *c*(*i*, *j*) is the cost of assigning task *i* to worker *j* (*i*, *j* = 1, 2)
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    - Each worker is assigned to exactly one task
    - The total cost is minimal

	1	**2**
1	4	2
2	7	1

- **Modelling**
  - Variable definition
    - Binary variable $X(i, j)$ in which $X(i, j) = 1$ means that task *i* is assigned to worker *j*, and $X(i, j) = 0$ means that task *i* is not assigned to worker *j* ($i, j = 1, 2$)
  - Constraints
    - $X(1, 1) + X(1, 2) = 1$; $X(2, 1) + X(2, 2) = 1$
    - $X(1, 1) + X(2, 1) = 1$; $X(1, 2) + X(2, 2) = 1$
  - Objectives
    - Minimize $f(X) = 4X(1, 1) + 2X(1, 2) + 7X(2, 1) + X(2, 2)$

SOFTWARE TOOLS

- CPLEX
- Gurobi
- SCIP
- OR-Tools

<table>
<tr><th>Mathematical model</th><th>OR-Tools</th></tr>
<tr><td>

- Variable definition
  - X_1, X_2: amount of P1 and P2 to be produced
  - Domains: $X_1, X_2 \in \mathbb{R}$ and nonnegative
- Constraints
  - $5X_1 + 3X_2 \leq 200$
  - $2X_1 + 3X_2 \leq 150$
- Objectives
  - Maximize $5X_1 + 7X_2$

</td><td>

```
from ortools.linear_solver import pywraplp

solver = pywraplp.Solver.CreateSolver("GLOP")
X1 = solver.NumVar(0, solver.infinity(), "X1")
X2 = solver.NumVar(0, solver.infinity(), "X2")

solver.Add(5*X1 + 3*X2 <= 200)
solver.Add(2*X1 + 3*X2 <= 150)

solver.Maximize(5 * X1 + 7 * X2)

status = solver.Solve()
if status == pywraplp.Solver.OPTIMAL:
 print("Solution:")
 print(f"Objective value = {solver.Objective().Value():0.1f}")
 print(f"x = {X1.solution_value():0.1f}")
 print(f"y = {X2.solution_value():0.1f}")
else:
 print("The problem does not have an optimal solution.")
```

</td></tr>
</table>

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THANK YOU !